

Secrets to a Good Password

Any level of encryption is useless unless secured by a strong key. The following guides may seem like common sense, but very few remember to adhere to them.

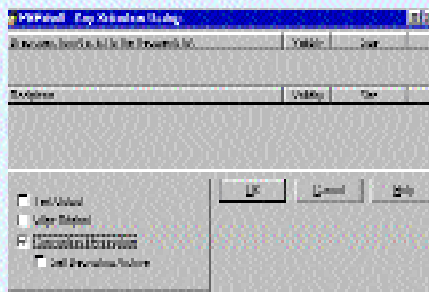
Firstly, don't stick your password on your monitor and don't give it out to anyone else, no matter how much you trust them. You then need a password that is difficult to guess or crack. The best passwords are a combination of eight or more upper and lower case letters, numbers and symbols. Don't choose a common phrase and never use the same password as your login. Most fraud cases are inside jobs, so names and birth dates of loved ones are out as well.

Apart from names, nicknames and dates, the most commonly used passwords are god, sex, password, angel, trustme, one's address or hometown and complicated-looking, but easy-to-remember key combinations such as z1x2c3v4. The characters may be upper or lower case, but don't use any of these for passwords.

It is also important to vary your passwords. Having the same password for everything is dangerous—if one password is discovered, all your files and accounts are at risk. Also, change your passwords on a regular basis, but don't use a standard policy to build your passwords—change the strategy too.

Passwords are normally cracked in three ways. The most common is guessing. Names, or combinations of names and important dates have often given way past all security schemes. Second is a dictionary attack, where words and word combinations from a dictionary of commonly used names and words end up forming the right password. Thus, don't use common words and phrases. Lastly, the brute force attack tries every possible combination of letters, numbers and symbols to find the right password. The more complicated the password, the longer it will take to crack. In fact, some good passwords have been estimated to take years to crack on very high-end systems.

A computer-generated random password is usually very good, but difficult to remember. There are several software available that can maintain password lists in an encrypted file, so you would need to remember just the password to access this list.



Encrypting files on your PC using PGP

works, keeping your key to yourself is of no use. Those who need to send you secure messages need to use your public key to encrypt the message. Run PGPKeys from the PGP system tray icon, right-click your key and click Export. It will be saved as an ASC file and can be distributed to others. Be careful to never send out your private key. Similarly, to be able to send secure messages, you will need the recipients' public keys. Import the key from PGPKeys and you are ready to send encrypted messages. When using Outlook or Outlook Express, click **Tools > Encrypt using PGP**.

To encrypt files, right-click the file in Windows Explorer and click **PGP > Encrypt**. PGP will bring up a window where you can set encryption parameters. Choose Conventional Encryption and click OK. You will be asked for a passphrase for encrypting the file, after which the file will be secured. The file can also be decrypted by right-clicking from Windows Explorer.

Don't let the bugs bite

Viruses can wipe out all data on your PC or corrupt the files internally, making recovery impossible. In a networked environment, they can be even more dangerous.

Anti-virus software are aplenty today and many are very effective in controlling deadly viruses. When choosing an anti-virus software, your primary concern should be effectiveness. It should be able to reliably trap all known viruses and prevent them from infecting your files and applications. It should also transparently monitor all activity on your computer and check every new file that's coming in. A good virus scanner understands the nature of viruses and prevents not only known viruses, but also identifies viral behaviour so that it can detect new, as yet unknown viruses.

Finally, the software vendor should provide regular updates for new viruses. This is critical to securing your system as new viruses are being let loose every day. Alerts of new viruses via e-mail is an added

bonus—there is always some time between the time the virus is released and a cure is developed. If the vendor can inform you of the virus beforehand with time for avoiding it, you can secure your system accordingly. Ensure that the anti-virus software is always running and monitoring your system. With the increasing penetration of the Internet, and the changing nature and attack methods of viruses, security is very important.

AVG Anti-Virus (www.grisoft.com) is a pretty good free virus scanner. It continuously monitors all disks and even e-mails that your e-mail is safe. The developers also provide regular updates that you can download from their Web site.

Great walls of fire

A standard advice from professional security is to not connect your PC to any network, even the Internet, and never share any files! This is too extreme and unpractical, but it does highlight the fact that as long as you are connected, you are vulnerable to viruses and hackers. Just like you use an anti-virus software to prevent viruses, you can use a firewall for monitoring network traffic.

At the most basic level, a firewall separates an untrusted network such as the Internet from a trusted network connected to your PC. The firewall works like a bouncer or a gatekeeper who directs visitors to the correct people and makes sure that intruders do not hinder the work. A good firewall will analyse all network traffic and allow only authorised data to pass through. Also, the firewall itself must be immune to viruses and intrusion. It may also log network information for user analysis.

Many types of firewalls exist as software and hardware. Hardware firewalls are usually implemented in dedicated organisations. Software firewalls function just like proxy servers, but instead of simply connecting networks, they also analyse the data packets passing through their origin as well as the intended destination. These firewalls also provide detailed reports than hardware firewalls.

Firewalls can prevent Denial of Service (DoS) attacks to your PC or network. Though it is fairly easy for pranksters to find out your IP address (this is easier on broadband connections where you could have fixed IPs) and send off a flood of attack, most such attacks are a result of random practices, without any particular machine in mind. On an unprotected system, a DoS attack can freeze the OS,